Multi-Agent Game Theory at Scale: Mathematical Foundations for Agentic AI

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Abstract

These notes provide a self-contained treatment of multi-agent game theory as it applies to modern large-scale AI systems. We develop the mathematical foundations of Nash equilibria, counterfactual regret minimisation (CFR), and mean-field games, then trace the line from classical solution concepts through landmark systems—Libratus, Pluribus, Cicero—to contemporary questions about multi-agent LLM deployments and AI safety. The target audience is Master's students with background in machine learning, optimisation, and probability theory.

Contents

1 Introduction

Game theory studies strategic interaction: settings where the outcome for each participant depends on the actions of all. As AI systems move from single-agent pipelines to *agentic* architectures—where multiple LLM-powered agents negotiate, plan, and act in shared environments—the classical toolkit of Nash equilibria, regret minimisation, and mechanism design becomes directly relevant.

Why game theory for ML researchers.

(i) **Training as a game.** GANs, RLHF, and self-play all involve multiple interacting objectives.

(ii) **Deployment as a game.** Autonomous agents bidding in auctions, negotiating contracts, or coordinating in multi-agent environments face genuine strategic uncertainty.

(iii) **Safety as a game.** Alignment can be formalised as a principal–agent problem; robustness against adversarial attacks is a zero-sum game.

The notes are organised as follows. Section **??** reviews classical solution concepts. Section **??** develops counterfactual regret minimisation. Sections **??**–**??** cover the landmark poker and Diplomacy AI systems. Section **??** introduces mean-field games. Section **??** treats opponent shaping. Section **??** draws connections to modern agentic AI. Section **??** lists open problems, and Section **??** provides an annotated bibliography.

2 Nash Equilibria and Classical Solution Concepts

2.1 Normal-Form Games

Definition 2.1 (Normal-form game)**.** A *finite normal-form game* is a tuple $\Gamma = (N, (A_i)_{i\in N}, (u_i)_{i\in N})$ where

- $N = \{1, \dots, n\}$ is the set of *players*,
- A_i is the finite *action set* of player i,
- $u_i : \prod_j A_j \to \mathbb{R}$ is the *utility function* of player i.

A *mixed strategy* for player i is a distribution $\sigma_i \in \Delta(A_i)$. Write $\sigma = (\sigma_i)_{i\in N}$ for a *strategy profile* and σ_{-i} for the profile of all players except i. The expected utility under σ is

$$U_i(\sigma) = \sum_{a\in\prod_j A_j} \Big(\prod_{j\in N} \sigma_j(a_j)\Big)\, u_i(a).$$

2.2 Nash Equilibrium

Definition 2.2 (Nash equilibrium)**.** A strategy profile σ^* is a *Nash equilibrium* (NE) if, for every player i and every alternative strategy $\sigma_i' \in \Delta(A_i)$,

$$U_i(\sigma_i^*, \sigma_{-i}^*) \;\geq\; U_i(\sigma_i', \sigma_{-i}^*).$$

Equivalently, every player plays a *best response* to the others.

Definition 2.3 (ε-Nash equilibrium)**.** A profile σ is an ε-Nash equilibrium if for every player i and every σ_i',

$$U_i(\sigma_i, \sigma_{-i}) \;\geq\; U_i(\sigma_i', \sigma_{-i}) - \varepsilon.$$

2.3 Nash's Existence Theorem

Theorem 2.4 (Nash, 1950)**.** *Every finite normal-form game possesses at least one mixed-strategy Nash equilibrium.*

Proof sketch via Kakutani's fixed-point theorem. Define the *best-response correspondence* $\mathrm{BR}_i(\sigma_{-i}) = \arg\max_{\sigma_i' \in \Delta(A_i)} U_i(\sigma_i', \sigma_{-i})$. Let $F : \Sigma \rightrightarrows \Sigma$ be the joint best-response map $F(\sigma) = \prod_i \mathrm{BR}_i(\sigma_{-i})$ where $\Sigma = \prod_i \Delta(A_i)$. The set Σ is a non-empty, compact, convex subset of Euclidean space. Each BR_i is upper-hemicontinuous (by the Maximum Theorem) and convex-valued (since U_i is linear in σ_i). Hence F satisfies the hypotheses of *Kakutani's fixed-point theorem*, so there exists $\sigma^* \in F(\sigma^*)$, which is by construction a Nash equilibrium. □

2.4 Computational Complexity of Nash Equilibria

Theorem 2.5 (Daskalakis, Goldberg, Papadimitriou, 2009; Chen, Deng, 2006)**.** *Computing a Nash equilibrium of a two-player normal-form game is* PPAD-complete.

Remark 2.6. PPAD (Polynomial Parity Arguments on Directed graphs) is a complexity class that captures the difficulty of problems whose existence is guaranteed by a parity argument. PPAD-completeness implies that, unless PPAD $\subseteq$ P (widely believed false), there is no polynomial-time algorithm for finding an exact NE. This motivates the search for *approximate* equilibria and equilibria of *structured* game classes.

2.5 Correlated Equilibrium

Definition 2.7 (Correlated equilibrium, Aumann 1974)**.** A distribution $\mu \in \Delta(\prod_i A_i)$ is a *correlated equilibrium* (CE) if, for every player i and every function $\phi_i : A_i \to A_i$,

$$\sum_{a \in \prod_j A_j} \mu(a)\, u_i(a) \;\geq\; \sum_{a \in \prod_j A_j} \mu(a)\, u_i(\phi_i(a_i), a_{-i}).$$

The set of correlated equilibria is a *convex polytope* defined by linear inequalities, so a CE can be found in polynomial time via linear programming. Every NE induces a CE (take $\mu = \prod_i \sigma_i^*$), but the converse is false; the CE polytope is generally larger.

2.6 Coarse Correlated Equilibrium and No-Regret Learning

Definition 2.8 (Coarse correlated equilibrium)**.** A distribution μ is a *coarse correlated equilibrium* (CCE) if, for every player i and every action $a_i' \in A_i$,

$$\mathbb{E}_{a \sim \mu}[u_i(a)] \;\geq\; \mathbb{E}_{a \sim \mu}[u_i(a_i', a_{-i})].$$

The connection to learning is fundamental:

Theorem 2.9 (No-regret $\Rightarrow$ CCE convergence)**.** *If every player i uses a no-external-regret learning algorithm (so that each player's regret R_i^T satisfies $R_i^T/T \to 0$), then the empirical distribution of play converges to the set of coarse correlated equilibria.*

Proof sketch. Let $\bar{\mu}^T = \frac{1}{T}\sum_{t=1}^T \delta_{a^t}$ be the empirical distribution. Player i's regret is $R_i^T = \max_{a_i'} \sum_{t=1}^T [u_i(a_i', a_{-i}^t) - u_i(a^t)]$. Dividing by T and rearranging gives exactly the CCE condition with slack $R_i^T/T \to 0$. □

Remark 2.10. If every player uses a no-*internal*-regret (or no-*swap*-regret) algorithm, the empirical play converges to the set of *correlated* equilibria. In two-player zero-sum games, no-regret learning yields convergence to Nash equilibrium of the *average* strategies (though not the last iterate in general).

3 Counterfactual Regret Minimisation (CFR)

3.1 Extensive-Form Games

Definition 3.1 (Extensive-form game of imperfect information)**.** An extensive-form game is specified by:

- A game tree with *histories* $\mathcal{H}$, terminal histories $\mathcal{Z} \subset \mathcal{H}$, and a player function $P : \mathcal{H} \setminus \mathcal{Z} \to N \cup \{c\}$ (where c denotes chance).
- For each player i, an *information partition* $\mathcal{I}_i$ that groups histories into *information sets*; at histories in the same information set, the player cannot distinguish the game states.
- Action sets $A(I)$ for each information set $I \in \mathcal{I}_i$, with $A(h) = A(I)$ for all $h \in I$.
- Utility functions $u_i : \mathcal{Z} \to \mathbb{R}$.

A *behavioural strategy* σ_i assigns a distribution over $A(I)$ to each information set $I \in \mathcal{I}_i$. By Kuhn's theorem, in games of perfect recall, behavioural and mixed strategies are equivalent.

Notation. For a terminal history z and strategy profile σ, write $\pi^\sigma(z) = \prod_{h \sqsubset z} \sigma_{P(h)}(a_h)$ for the probability of reaching z, where a_h is the action taken at h on the path to z. Decompose:

$$\pi^\sigma(z) = \pi_i^\sigma(z) \cdot \pi_{-i}^\sigma(z) \cdot \pi_c(z),$$

where $\pi_i^\sigma(z)$ is the product of player i's action probabilities, π_{-i}^σ is the opponents', and π_c is chance's.

3.2 Counterfactual Value

Definition 3.2 (Counterfactual value)**.** The *counterfactual value* of information set $I \in \mathcal{I}_i$ under profile σ is

$$v_i^\sigma(I) = \sum_{z \in \mathcal{Z}_I} \pi_{-i}^\sigma(z[I])\, \pi_c(z[I])\, \pi^\sigma(z \mid z[I])\, u_i(z),$$

where $\mathcal{Z}_I$ is the set of terminal histories passing through I, $z[I]$ is the prefix of z at I, and $\pi^\sigma(z \mid z[I])$ is the probability of the continuation from $z[I]$ to z under σ. Intuitively, this is the expected value for player i at I, weighted by the opponents' and chance's probability of reaching I, but *assuming player i plays to reach I*.

For a specific action $a \in A(I)$:

$$v_i^\sigma(I, a) = \sum_{z \in \mathcal{Z}_{I,a}} \pi_{-i}^\sigma(z[I])\, \pi_c(z[I])\, \pi^\sigma(z \mid z[I], a)\, u_i(z).$$

3.3 Counterfactual Regret and Regret Matching

Definition 3.3 (Instantaneous counterfactual regret)**.** For action a at information set I at iteration t:

$$r^t(I, a) = v_i^{\sigma^t}(I, a) - v_i^{\sigma^t}(I).$$

The *cumulative counterfactual regret* is $R^T(I, a) = \sum_{t=1}^T r^t(I, a)$.

Algorithm 1 Vanilla CFR

Initialise σ^1 uniformly, $R^0(I,a) \leftarrow 0$, $S^0(I,a) \leftarrow 0$ for all I, a.
for $t = 1, 2, \ldots, T$ **do**
  for each player i **do**
    Traverse game tree; compute $v_i^{\sigma^t}(I,a)$ for all $I \in \mathcal{I}_i$, $a \in A(I)$.
    $R^t(I,a) \leftarrow R^{t-1}(I,a) + r^t(I,a)$
    $S^t(I,a) \leftarrow S^{t-1}(I,a) + \pi_i^{\sigma^t}(I)\,\sigma^t(I,a)$ ▷ Average strategy weights
  end for
  Update σ^{t+1} via regret matching.
end for
return $\bar{\sigma}^T(I,a) = S^T(I,a) / \sum_{a'} S^T(I,a')$ ▷ Average strategy

Regret matching. At each iteration, CFR updates the strategy at each information set via *regret matching*:

$$\sigma^{T+1}(I,a) = \begin{cases} \dfrac{R^{T,+}(I,a)}{\sum\limits_{a'} R^{T,+}(I,a')} & \text{if } \sum_{a'} R^{T,+}(I,a') > 0, \\ \dfrac{1}{|A(I)|} & \text{otherwise,} \end{cases}$$

where $R^{T,+}(I,a) = \max(R^T(I,a), 0)$.

3.4 Convergence Theorem

Theorem 3.4 (CFR convergence, Zinkevich et al. 2007)**.** *In a two-player zero-sum extensive-form game of perfect recall, the average strategy profile $\bar{\sigma}^T$ produced by CFR is a $2\Delta_u\sqrt{|\mathcal{I}| \cdot \max_I |A(I)|} \,/\, \sqrt{T}$-Nash equilibrium, where $\Delta_u = \max_z u(z) - \min_z u(z)$ is the range of utilities and $|\mathcal{I}| = |\mathcal{I}_1| + |\mathcal{I}_2|$.*

Proof sketch. The key insight is the *counterfactual regret decomposition*: the overall regret of player i is bounded by the sum of positive counterfactual regrets:

$$R_i^T \;\le\; \sum_{I \in \mathcal{I}_i} R_{\max}^{T,+}(I), \qquad R_{\max}^{T,+}(I) = \max_a R^{T,+}(I,a).$$

Regret matching at each information set guarantees $R_{\max}^{T,+}(I) \le \Delta_u\sqrt{|A(I)| \cdot T}$ (this follows from the regret bound of the multiplicative-weights / regret-matching algorithm). Therefore:

$$R_i^T \le \sum_{I \in \mathcal{I}_i} \Delta_u \sqrt{|A(I)|\,T} \le \Delta_u \sqrt{T} \cdot |\mathcal{I}_i| \cdot \sqrt{\max_I |A(I)|}.$$

In a two-player zero-sum game, if both players have average regret at most ε, then the average strategy profile is a 2ε-Nash equilibrium (since the game value is well-defined and each player's average strategy is ε-close to a best response). The bound $\varepsilon = O(1/\sqrt{T})$ follows. □

3.5 CFR+ and Variants

CFR+ (Tammelin et al., 2015; Bowling et al.) modifies the regret accumulation rule to

$$R^t(I,a) \leftarrow \max\bigl(R^{t-1}(I,a) + r^t(I,a),\ 0\bigr),$$

i.e., cumulative regrets are clipped to non-negative values at every step ("regret matching+"). Additionally, the average strategy uses linear weighting $S^t(I,a) \leftarrow S^{t-1}(I,a) + t \cdot \pi_i^{\sigma^t}(I)\,\sigma^t(I,a)$,

which upweights later (better) strategies. Empirically, CFR+ converges orders of magnitude faster than vanilla CFR.

Monte Carlo CFR (MCCFR) uses sampling to avoid full tree traversals. Variants include *outcome sampling* (single trajectory) and *external sampling* (sample opponent/chance, traverse all of player i's actions).

Deep CFR (Brown et al., 2019) replaces the regret tables with neural networks. At each CFR iteration, advantage values are computed via traversals and stored in a reservoir buffer; a neural network is trained to predict cumulative advantages $\hat{R}^T(I, a)$. A separate network approximates the average strategy.

4 Libratus and Pluribus

4.1 Libratus: Superhuman Heads-Up No-Limit Texas Hold'em

Libratus (Brown and Sandholm, 2018) defeated top professional players in heads-up no-limit Texas Hold'em. The system has three modules:

1. **Blueprint computation.** An abstracted version of the game is solved using Monte Carlo CFR. Card abstraction groups strategically similar hands; bet abstraction discretises the continuous betting actions.

2. **Subgame solving.** During play, when the opponent takes an action outside the abstraction, Libratus constructs and solves a *subgame* in real time.

3. **Self-improver.** After each day of play, Libratus adds new bet sizes to the abstraction at branches the opponents frequently visited, then re-solves.

4.2 Safe Subgame Solving

Naive subgame solving can increase exploitability. The key theoretical contribution is *safe* (or reach-aware) subgame solving:

Theorem 4.1 (Safe subgame solving, Brown and Sandholm 2017)**.** *Let σ be a Nash equilibrium of the full game and let S be a subgame rooted at a public state. If we re-solve S subject to the constraint that the value achieved for each opponent reach distribution is at least as high as under σ, then the resulting strategy is part of a Nash equilibrium of the full game.*

Remark 4.2. In practice, one does not have an NE of the full game; one has the blueprint $\hat{\sigma}$. Subgame solving refines the blueprint locally, and the "safe" constraints prevent any increase in exploitability relative to $\hat{\sigma}$.

4.3 Pluribus: Six-Player No-Limit Hold'em

Pluribus (Brown and Sandholm, 2019) extended superhuman play to six-player poker. Key innovations:

- **Depth-limited search.** Instead of solving to the end of the game, Pluribus searches only a few moves ahead and uses a learned *value function* (the blueprint strategy's expected value) at leaf nodes.

- **Linear CFR.** Regrets are weighted linearly by iteration number: $R^T(I, a) = \sum_{t=1}^{T} t \cdot r^t(I, a)$, discounting early (poor) iterations.

- **Multi-player subgame solving.** In multiplayer games, Nash equilibria are not unique and not interchangeable. Pluribus handles this by assuming opponents play according to a small number of "blueprint-like" strategies and solving against this restricted opponent model.

5 Cicero: AI for Diplomacy

5.1 The Challenge of Diplomacy

Diplomacy is a seven-player board game of negotiation and alliance formation. Unlike poker, it requires:

- **Natural language communication** — players negotiate via free-form messages.
- **Cooperation and betrayal** — alliances are non-binding.
- **Very large action spaces** — each turn involves simultaneous orders for multiple units.

No prior AI had achieved human-level play.

5.2 Cicero's Architecture (Meta FAIR, 2022)

Cicero (FAIR/Meta, led by Noam Brown et al.) integrates:

1. **Strategic reasoning engine.** A planning module based on *piKL* (policy-regularised iterative Kullback–Leibler divergence). Given a prior policy π_0 (a supervised-learning model trained on human games), piKL solves:

$$\sigma_i^* = \arg\max_{\sigma_i} \left[U_i(\sigma_i, \sigma_{-i}) - \lambda \, \mathrm{KL}(\sigma_i \| \pi_0) \right],$$

   iterating best responses while staying close to human-like play. This is crucial: pure equilibrium play would be alien to human partners and undermine cooperation.

2. **Dialogue model.** A large language model (fine-tuned BART/R2C2) that generates natural language messages *conditioned on the intended actions.*

3. **Intent model.** Predicts other players' intended actions from their messages, enabling Cicero to "read between the lines."

4. **Filtering pipeline.** Generated messages are filtered for consistency with the strategic plan, honesty (relative to Cicero's actual intentions), and coherence.

5.3 Signaling Games Formalization

Diplomacy can be partially formalised through the lens of *signaling games* (Crawford and Sobel, 1982):

Definition 5.1 (Signaling game)**.** A signaling game consists of a *sender* with private type $\theta \in \Theta$, a *message space* M, and a *receiver* who takes action $a \in A$ after observing the message. A *perfect Bayesian equilibrium* specifies:

- Sender strategy $\mu : \Theta \to \Delta(M)$,
- Receiver strategy $\alpha : M \to \Delta(A)$,
- Receiver beliefs $\beta : M \to \Delta(\Theta)$ consistent with Bayes' rule on the equilibrium path.

In Diplomacy, each player is simultaneously a sender and receiver. The "type" corresponds to the player's strategic intent (which units to move where), the messages are natural language, and the actions are the orders submitted. Cicero's architecture approximates an equilibrium of this signaling game: the intent model estimates β, the strategic engine computes α (best-response actions given beliefs), and the dialogue model implements μ (generating messages consistent with intent).

A key subtlety is *cheap talk* (messages are costless): in classical cheap talk games, only "babbling" equilibria survive under misaligned incentives. Diplomacy partly escapes this because players have *partially aligned* interests (they need allies) and the game is repeated (reputation matters).

6 Mean-Field Games

6.1 Motivation: The $N \to \infty$ Limit

Many real-world multi-agent systems involve a very large number of interacting agents: financial markets, autonomous vehicle fleets, content recommendation ecosystems. Solving for Nash equilibria in N-player games is intractable for large N. *Mean-field game theory* (independently introduced by Lasry and Lions, 2006–2007, and Huang, Malhamé, and Caines, 2006) studies the limiting regime $N \to \infty$, where each agent interacts with the *population distribution* rather than with individual agents.

6.2 The Mean-Field Game PDE System

Consider a continuum of agents, each with state $x \in \mathbb{R}^d$, evolving according to the controlled stochastic differential equation:

$$dX_t = b(X_t, \alpha_t, m_t)\, dt + \sigma\, dW_t,$$

where α_t is the control, m_t is the population distribution at time t, and W_t is a standard Brownian motion. Each agent minimises a cost functional:

$$J(\alpha; m) = \mathbb{E}\left[\int_0^T L(X_t, \alpha_t, m_t)\, dt + G(X_T, m_T)\right].$$

A *mean-field Nash equilibrium* is a pair (α^*, m^*) such that:

1. α^* is optimal for the cost $J(\cdot\,; m^*)$ (taking m^* as given),
2. m^* is the distribution of the state process under α^*.

This fixed-point condition yields a coupled PDE system:

Theorem 6.1 (Mean-field game PDE system, Lasry and Lions 2007)**.** *Under regularity conditions, a mean-field Nash equilibrium is characterised by the system*

$$-\partial_t u - \nu \Delta u + H(x, \nabla u, m) = 0 \qquad \textit{(Hamilton–Jacobi–Bellman, backward)}, \tag{1}$$

$$\partial_t m - \nu \Delta m - \operatorname{div}(m\, \nabla_p H(x, \nabla u, m)) = 0 \qquad \textit{(Fokker–Planck / Kolmogorov, forward)}, \tag{2}$$

with terminal condition $u(T, x) = G(x, m_T)$ and initial condition $m(0, \cdot) = m_0$, where H is the Hamiltonian of the optimal control problem and $\nu = \sigma^2/2$.

The HJB equation (**??**) runs *backward* in time from the terminal cost, determining the value function and hence the optimal policy. The Fokker–Planck equation (**??**) runs *forward*, propagating the population density under the optimal policy. The coupling makes this a challenging forward–backward system.

6.3 Existence and Uniqueness: Lasry–Lions Monotonicity

Definition 6.2 (Lasry–Lions monotonicity condition)**.** The running cost L and terminal cost G satisfy the *Lasry–Lions monotonicity condition* if, for all distributions m_1, m_2,

$$\int_{\mathbb{R}^d} \big[F(x, m_1) - F(x, m_2)\big]\, d(m_1 - m_2)(x) \;\geq\; 0,$$

where F denotes the coupling terms through which m enters the costs.

Theorem 6.3 (Uniqueness under monotonicity)**.** *If the Lasry–Lions monotonicity condition holds strictly, then the mean-field game PDE system has at most one solution, and hence the mean-field Nash equilibrium is unique.*

Intuitively, monotonicity means that agents are "repelled" by crowding—costs increase where the population concentrates. This is natural in congestion games but violated in models with herding or network effects.

6.4 Mean-Field Multi-Agent Reinforcement Learning

In the ML setting, the MFG framework is combined with RL:

- **Mean-Field MARL** (Yang et al., 2018): each agent's Q-function depends on its own action and the mean action of neighbours, reducing the joint action space from $|A|^N$ to $|A| \times |A|$.
- **MF-OMO** (Gemp et al., DeepMind): online mirror descent in the space of population distributions.
- **Neural MFG** (Carmona and Laurière, 2019): solve the coupled PDE system using neural networks (physics-informed or via direct policy optimisation).

7 Opponent Shaping

7.1 Beyond Best Response: Shaping Opponents' Learning

Standard game-theoretic algorithms assume opponents are fixed or adversarial. *Opponent shaping* considers agents that explicitly account for their influence on other agents' learning dynamics.

7.2 LOLA: Learning with Opponent-Learning Awareness

Foerster et al. (2018) introduced LOLA. Consider two agents with parameters θ_1, θ_2 and differentiable expected returns $V_1(\theta_1, \theta_2)$, $V_2(\theta_1, \theta_2)$. A naive learner updates:

$$\theta_1 \leftarrow \theta_1 + \alpha\, \nabla_{\theta_1} V_1(\theta_1, \theta_2).$$

LOLA anticipates that player 2 will perform a gradient step, so player 1 optimises V_1 *after* player 2's anticipated update:

Definition 7.1 (LOLA update)**.** Player 1's LOLA gradient is:

$$\nabla_{\theta_1}^{\mathrm{LOLA}} V_1 = \nabla_{\theta_1} V_1 + \alpha_2\, \nabla_{\theta_1} \big(\nabla_{\theta_2} V_2\big)^\top \nabla_{\theta_2} V_1,$$

where α_2 is player 2's learning rate. The additional term $\nabla_{\theta_1}(\nabla_{\theta_2} V_2)^\top \nabla_{\theta_2} V_1$ is a second-order correction that "steers" player 2's learning in a direction beneficial to player 1.

Remark 7.2. The LOLA update requires computing a Hessian–vector product $\nabla_{\theta_1}(\nabla_{\theta_2} V_2)$, which is tractable via automatic differentiation (forward-mode or reverse-mode with `jax.grad`). Higher-order extensions (LOLA-k, SOS) account for k levels of recursive reasoning.

Results. In the Iterated Prisoner's Dilemma, LOLA agents learn tit-for-tat–like strategies and achieve mutual cooperation, whereas naive learners converge to mutual defection.

7.3 Model-Free Opponent Shaping (M-FOS)

Lu et al. (2022) introduced M-FOS, which formulates opponent shaping as a *meta-game*: a "meta-agent" learns a policy over the space of inner-loop learning algorithms. The meta-agent's action at each step is a parameter update (or a choice of learning rule) for the inner agent.

Key features of M-FOS:

- Does not require differentiating through the opponent's update (model-free).
- Uses RL (PPO) to optimise the meta-agent's policy.
- Recovers LOLA-like behaviour in matrix games and scales to more complex settings.
- Can discover novel shaping strategies not captured by fixed-order Taylor approximations.

8 Connections to Modern Agentic AI

The recent proliferation of LLM-based agents creates new multi-agent settings where game-theoretic reasoning is essential.

8.1 Multi-Agent LLM Systems

When multiple LLM-powered agents interact—e.g., a buyer agent negotiating with a seller agent, or multiple coding agents working on a shared codebase—the interaction is inherently game-theoretic:

- Each agent has its own objective (its user's instructions).
- Agents may have private information (user preferences, budget constraints).
- Communication is in natural language, creating a signaling game.

Cicero's architecture provides a template: strategic reasoning (what to do) is separated from communication (what to say), with an intent model bridging the two.

8.2 Mechanism Design for AI

Mechanism design ("reverse game theory") asks: how should we design the rules of interaction to achieve desirable outcomes?

- **AI constitutions as mechanisms.** A system prompt or RLHF reward model defines the "rules" under which an LLM agent operates. Designing these to be incentive-compatible (so the agent has no reason to deviate) is a mechanism design problem.
- **Multi-agent coordination.** When multiple agents share a resource (API rate limits, shared memory, a code repository), a mechanism (e.g., a priority queue, a voting protocol) determines allocation. The VCG mechanism and its computational approximations are directly relevant.
- **Automated mechanism design.** Conitzer and Sandholm's line of work on computationally designing optimal mechanisms can be applied to design interaction protocols for AI agents.

8.3 Safety as Game Theory

- **Red-teaming as adversarial games.** Testing AI systems against adversarial attacks (jailbreaks, prompt injections) is a zero-sum game between attacker and defender.
- **Principal–agent models.** The alignment problem—ensuring an AI agent acts in its principal's interest—is a classical contract theory / principal–agent problem with moral hazard (hidden action) and adverse selection (hidden type).
- **Cooperative AI** (Dafoe et al., 2020). Designing agents that can credibly commit to cooperative strategies, even in the absence of binding agreements, requires game-theoretic tools (commitment devices, reputation, repeated game strategies).

8.4 Auction Theory for LLM APIs

LLM inference is a scarce resource. Multiple users compete for compute. This is naturally modelled as an auction:

- Users have private valuations for faster/better inference.
- The provider allocates compute (batch priority, model routing).
- Auction mechanisms (first-price, second-price, combinatorial) determine pricing and allocation.

The theoretical framework of Myerson (1981) optimal auction design applies. Recent work explores dynamic auctions for streaming compute, connecting to the bandit literature.

9 Open Problems

Open Problem 1 (Last-iterate convergence in general games)**.** No-regret algorithms converge in average to (coarse) correlated equilibria in general games and to Nash equilibria in two-player zero-sum games. Under what conditions do *last-iterate* convergence guarantees hold, and at what rate? Recent work (e.g., optimistic gradient descent/ascent) shows last-iterate convergence in zero-sum games, but the general-sum and multi-player cases remain largely open.

Open Problem 2 (Scalable equilibrium computation beyond two-player zero-sum)**.** CFR and its variants converge to NE in two-player zero-sum games. In general-sum or multiplayer games, NE need not be unique, may be non-interchangeable, and computing any NE is PPAD-hard. What practically relevant relaxations (e.g., team equilibria, extensive-form correlated equilibria, trembling-hand equilibria) are both tractable and strategically meaningful?

Open Problem 3 (Learning in mean-field games without monotonicity)**.** Most theoretical guarantees for MFG rely on the Lasry–Lions monotonicity condition. Real-world systems (social networks, financial markets) often exhibit herding, network effects, and strategic complementarities that violate monotonicity. Can we design provably convergent learning algorithms for non-monotone MFGs?

Open Problem 4 (Game-theoretic foundations of multi-agent LLM interaction)**.** When multiple LLM agents interact via natural language, what is the appropriate solution concept? Classical equilibria assume common knowledge of rationality, but LLM agents may have bounded rationality, prompt-dependent biases, and opaque objectives. Developing a formal theory—possibly drawing on bounded rationality, level-k thinking, or cognitive hierarchy models—is a major open problem.

Open Problem 5 (Mechanism design under LLM manipulation)**.** LLM-powered agents can generate persuasive but potentially deceptive text. Classical mechanism design assumes agents can misreport their *type* but cannot alter the mechanism itself. When agents can manipulate the mechanism designer (e.g., via prompt injection, social engineering), the revelation principle breaks down. How do we design robust mechanisms in this setting?

Open Problem 6 (Opponent shaping at scale)**.** LOLA and M-FOS have been demonstrated in small games. Scaling opponent shaping to large multi-agent environments (e.g., economy simulations, multi-agent robotics) with many heterogeneous agents is an open challenge. Connections to meta-learning and the mean-field limit may provide tractable approximations.

Open Problem 7 (Sample complexity of game solving with function approximation)**.** Deep CFR and similar methods use neural function approximation. What are the sample complexity guarantees? Can we characterise the approximation–regret trade-off when the counterfactual value function class has limited capacity?

10 Annotated Bibliography

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